

TECHNICAL ASSIGNMENT • 2026

FastBox Delivery System

A Logistics Simulation Built in Python



Python Assignment 2026

PROBLEM OVERVIEW

What We Had to Build



OBJECTIVE PIPELINE



Input Format — JSON

INPUT DATA

warehouses

locations • { x, y }

agents

workers • { id, location }

packages

deliveries • { warehouse, destination }

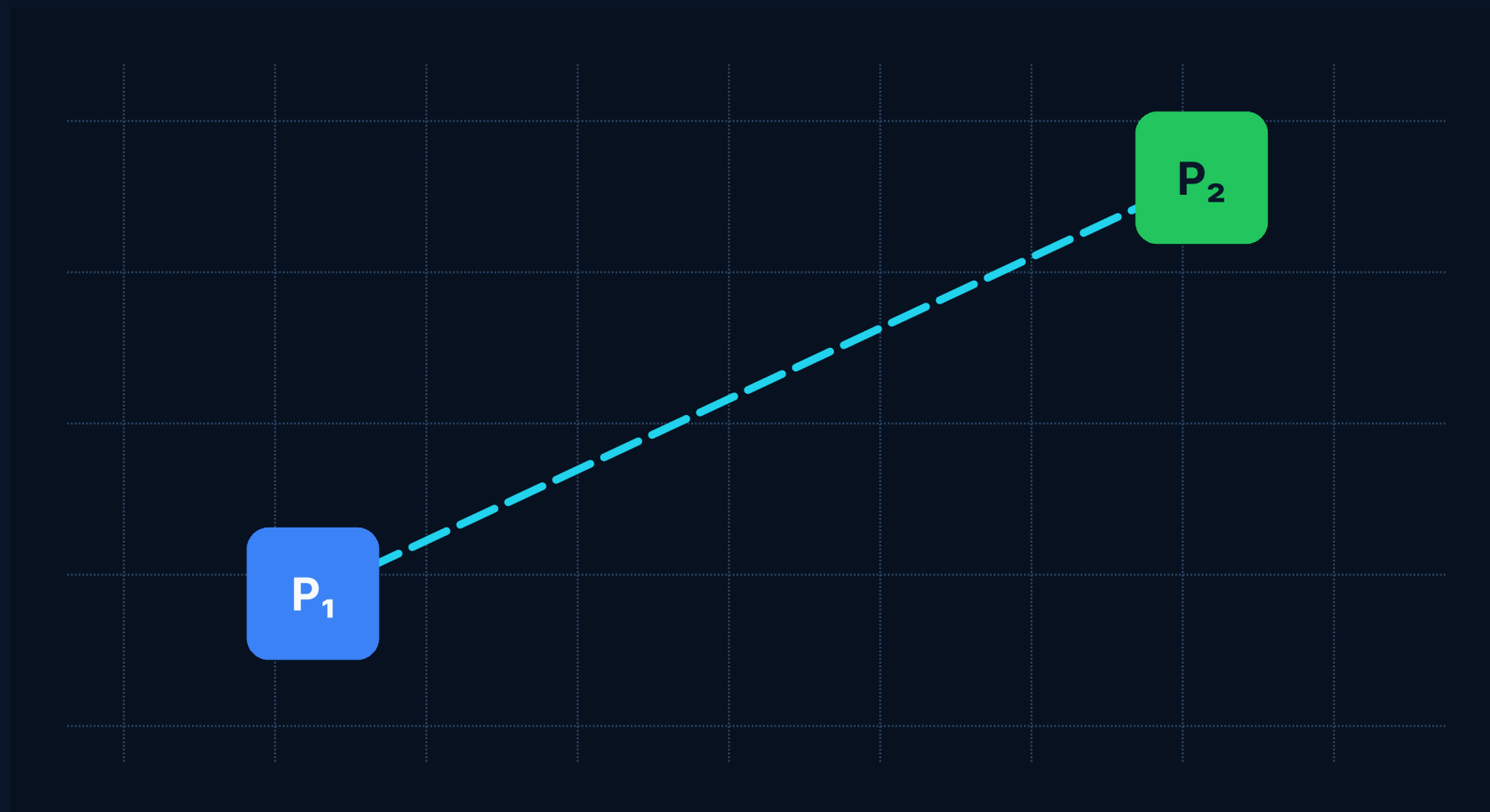
```
{
  "warehouses": [ /* placeholder data */ ],
  "agents":     [ /* placeholder data */ ],
  "packages":   [ /* placeholder data */ ]
}
```

How the Algorithm Works — Step by Step



Euclidean Distance — The Core Formula

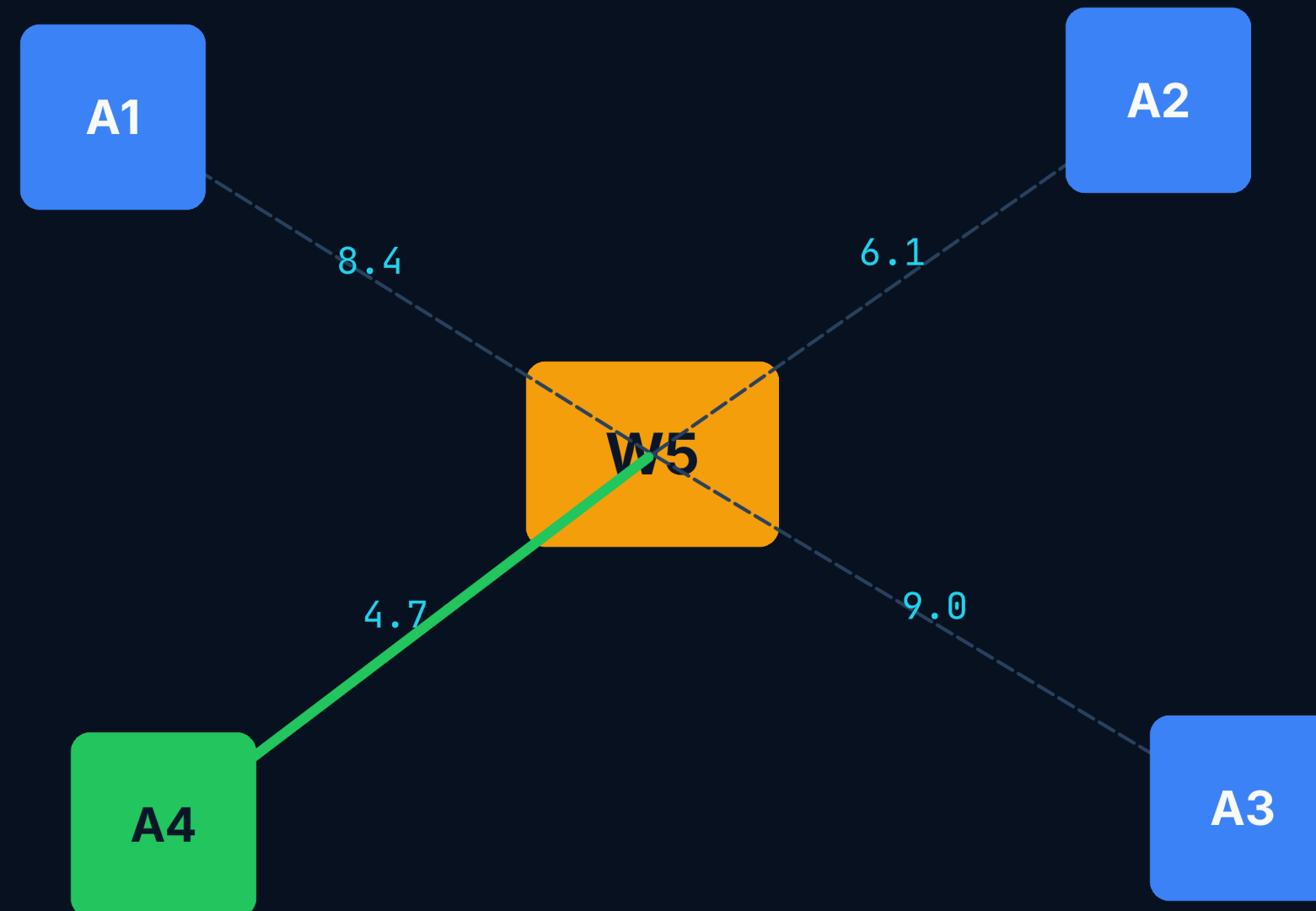
$$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$



```
def euclidean_distance(point_1, point_2):  
    # conceptual distance calculation  
    return distance
```

Distance between two points on a 2D grid

Assigning Packages to the Nearest Agent



1. For each package
2. Find its warehouse
3. Check all agents
- 4. Assign to the closest agent**
5. Apply alphabetical tie-breaking

Simulating the Delivery Journey

START POSITION

WAREHOUSE

DESTINATION

```
package_distance = leg1 + leg2
```

```
total_distance = sum of all packages
```

```
efficiency = total_distance ÷ packages_delivered
```

Agent position updates after each delivery.

Generated Report — report.json

```
{
  "agents": {
    "A1": { "packages_delivered": ...,
            "total_distance": ..., "efficiency": ... },
    "A2": { ... },
    "A3": { ... }
  },
  "best_agent": "..."
}
```

Best Agent

Lowest Efficiency Score
(least distance per package)

best_agent

Bonus Features Implemented

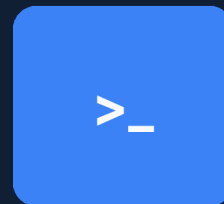
01



Random Delays

0–30 min per package,
reproducible with seed(42)

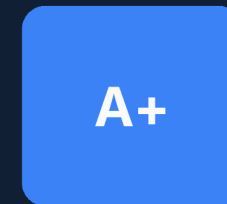
02



ASCII Route Map

Visual grid showing W / A / D
locations

03



New Agent Mid-Day

A_NEW joins at warehouse
centroid after 50% packages

04



CSV Export

Top performer saved to CSV
file

REPRESENTATIVE EXAMPLE

ASCII Route Map — Live Terminal Output

```
.....
...W.....
....A.....
.....D.....
.....W.....
.....A.....
.....
...D.....
.....D.....
.....W.....
.....A.....
.....
```

W = Warehouse | A = Agent | D = Destination

All coordinates normalised to grid; Y-axis inverted for correct orientation.

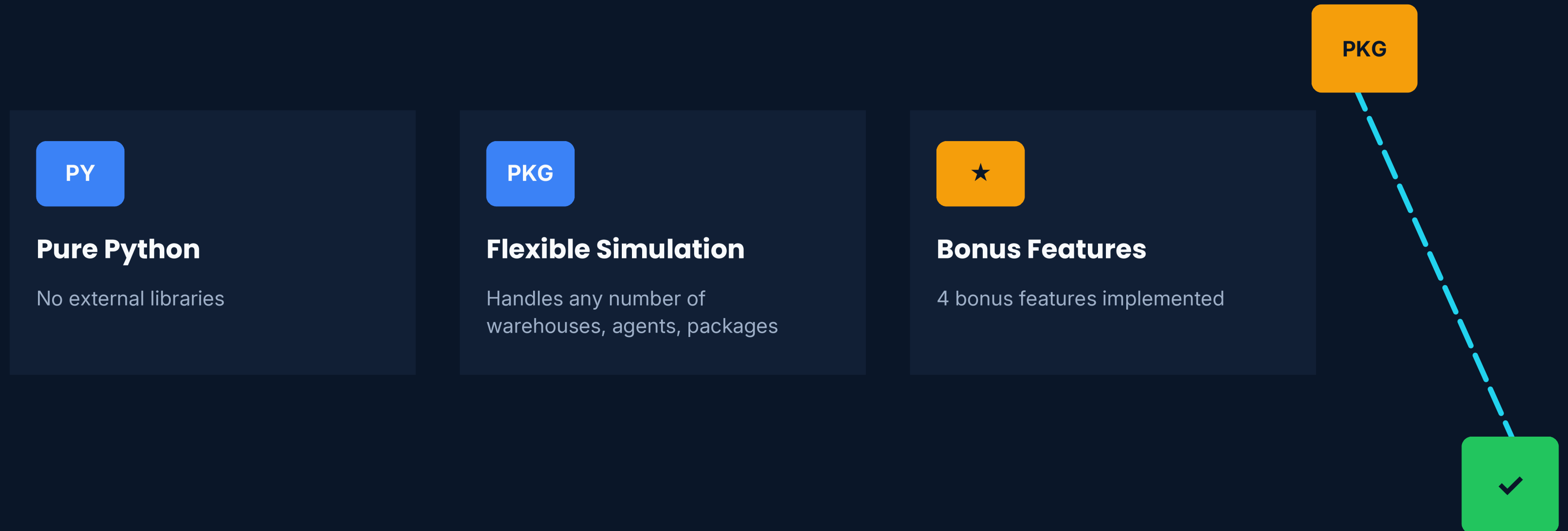
Test Results — All 11 Cases Passed

Test Case	Packages	Agents	Result
base_case	—	—	✓ PASS
TC1	—	—	✓ PASS
TC2	—	—	✓ PASS
TC3	—	—	✓ PASS
TC4	—	—	✓ PASS
TC5	—	—	✓ PASS
TC6	—	—	✓ PASS
TC7	—	—	✓ PASS
TC8	—	—	✓ PASS
TC9	—	—	✓ PASS
TC10	—	—	✓ PASS

total_delivered == total_packages

Zero packages lost.

Built to be clear, flexible, and verifiable.



Thank You